

Modeling of precipitation hardening during the aging and overaging of 18Ni–Co–Mo–Ti maraging 300 steel

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Abstract

The precipitation hardening behavior of an 18Ni–Co–Mo–Ti maraging 300 solution treated and aged in the 440–650 °C range was investigated. The aging at 440, 480, 510 and 560 °C was modeled by an equation $\Delta H = (Kt)^n$. The activation energy for precipitation was determined (132 kJ/mol) and compared to published values for 18Ni–Co–Mo–Ti maraging 250 and 350 steels. The model for overaging based on the Orowan mechanism was tested in samples treated at 560, 600 and 650 °C but was found to be valid only for 560 °C. The softening of samples aged at 600 and 650 °C is rather due to austenite precipitation than to Ni₃(Mo, Ti) and Fe₂Mo coarsening. Equations $H = H_1 t^p$ were fitted for the overaging at 560, 600 and 650 °C.

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1. Introduction

Maraging steels are age hardenable alloys for special purposes, where the combination of high strength and good toughness is required. The microstructure of the as quenched 18% maraging steel consists of low carbon and high nickel and cobalt martensite. The strengthening in low temperature aging (<450 °C) is produced by ordered and coherent phases such as μ , S and X phases [1–3].

The aging above 450 °C produces an intense hardening due to the precipitation of Ni₃(Mo, Ti) and Fe₂Mo phases [1–3].

Aging between 500 °C and the A_s temperature produces austenite precipitation by a diffusion controlled reaction [4–5]. According to Li and Yin [6] the austenite formation occurs at the same moment and as consequence of the partial dissolution of Ni₃(Ti, Mo) and precipitation of Fe₂Mo. Depending on the martensite start temperature (M_s), the austen-

ite formed by this way may be retained, partially or totally transformed into martensite during the cooling to room temperature. M_s decreases with the nickel content of the austenite, which is found to decrease with the aging temperature [7–8]. As a consequence, M_s increases with the aging temperature above 500 °C. The amount of austenite observed at room temperature increases with the aging temperature till a peak value and then decreases as the temperature M_s decreases. The maximum amount of reverted austenite observed at room temperature in 18Ni Co-containing maraging steels is obtained by aging treatments between 620 and 670 °C [9].

Wilson [10] presented and applied models for aging and overaging of precipitation hardenable alloys. Sha [11] applied these models for a maraging C-300 steel aged at 510 °C. The equation used for aging is based on the Johnson–Mehl and Avrami model [12–13]. The equation proposed for overaging is based on the Orowan mechanism. In the present work, these equations were tested for an 18Ni–Co–Mo–Ti maraging 300 steel aged in the 440–650 °C range.

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Table 1
Chemical composition of the maraging steel studied

wt. %	Element
0.01	C
17.86	Ni
9.41	Co
4.84	Mo
0.76	Ti
0.14	Al
0.01	Mn
balance	Fe

2. Experimental methods

A 2.5 mm thick sheet of maraging 300 steel (composition shown in Table 1) was solution treated at 900 °C for 40 min. After this the samples were aged at 440, 480, 510, 560 and 650 °C for different times from 15 min to 24 h. All heat treatments were conducted in vacuum-sealed quartz tubes.

Vickers hardness tests and X-ray diffraction analysis (XRD) were carried out in all conditions. All the X-ray measurements were carried out using a diffractometer PHILIPS®, model X'Pert Pro, in step scan mode with step size of 0.02° and time per step of 3 s. Cu K α (1.54056 Å) radiation was used with 40 kV and 40 mA. In order to keep the beam completely on the sample for low incident angles, a divergence slit of 1° was used.

3. Results

Fig. 1 shows the precipitation hardening curves of the maraging 300 steel aged at 440, 480, 510 and 560 °C. The aging curves at 440 and 480 °C do not present overaging till 24 h. The maximum hardness is obtained by the aging at 480 °C for 10 and 24 h. The samples aged at 510 °C achieve a maximum hardness of 629HV at 4 h of aging, and undergo a slight overaging after this.

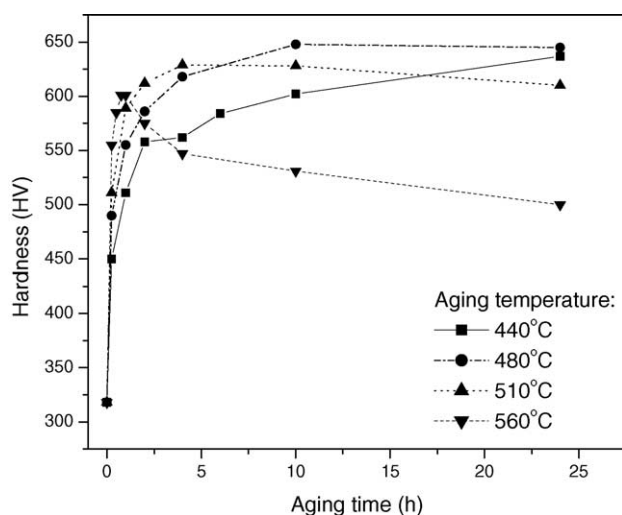


Fig. 1. Hardness vs. aging time at 440, 480, 510 and 560 °C.

Table 2
 n and K values at 440 °C, 480 °C, 510 °C and 560 °C

Temperature (°C)	Time range (h)	n	K (h $^{-1}$)	R
440	0.25–10	0.20 ± 0.02	1.3×10^{11}	0.97
480	0.25–2.0	0.20 ± 0.01	5.0×10^{11}	0.99
510	0.25–2.0	0.21 ± 0.02	3.9×10^{11}	0.99
560	0.25–0.75	0.16 ± 0.01	1.7×10^{15}	0.99

The model for kinetics of early stages of aging of precipitation hardenable alloys is given by an expression derived from the Avrami [13] equation [10]:

$$\Delta H = (Kt)^n \quad (1)$$

where K is the temperature dependent rate constant; n the time exponent, slightly temperature dependent; ΔH the increase of hardening (HV); t the aging time (h)

Table 2 shows the n and K values obtained by plotting $\ln(\Delta H)$ versus $\ln(t)$ (Fig. 2). The n values obtained at 440, 480 and 510 °C are close, but the one obtained at 560 °C is smaller. This is attributed to the softening effect caused by austenite precipitation at this temperature, even before the overaging. Fig. 3 shows the X-ray diffractogram of the sample aged at 560 °C by 45 min with austenite peaks.

The fitting for the 440 °C shown in Fig. 2 presented the lowest correlation coefficient ($R=0.97$). In this case, the points are better fitted by two curves (Fig. 4), where two aging stages are distinguished. The parameter n obtained for first and second stages were 0.29 ± 0.01 and 0.12 ± 0.01 , respectively. The different aging stages may be related to different types of precipitates in maraging steels.

The n and K values obtained at 510 °C in this work (Table 2) are somewhat different from the ones obtained by Sha [11] in a maraging steel of the same type. This can be attributed to differences in the time range used to fit the curves. In this work, we took points between 15 min and 2 h, while Sha [11] analyzed the hardening between 160 s and 25 min.

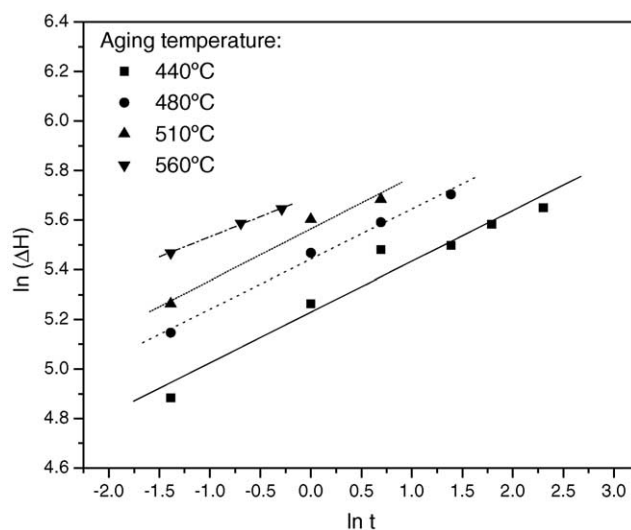


Fig. 2. $\ln(\Delta H)$ vs. $\ln(t)$ for the samples aged at 440, 480, 510 and 560 °C.

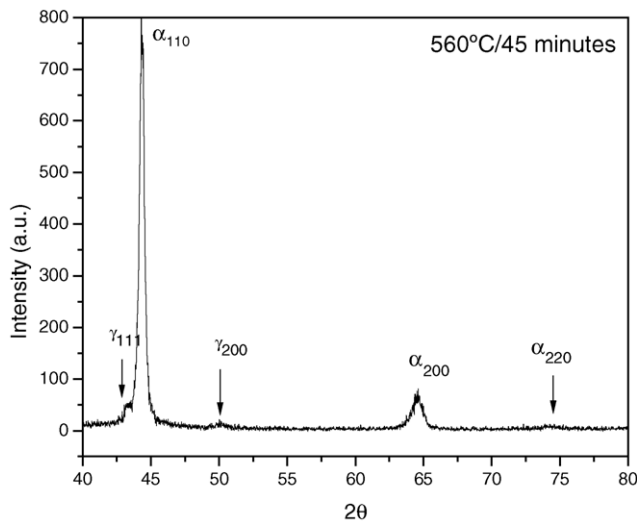


Fig. 3. X-ray diffractogram of the sample aged at 560 °C for 45 min.

The activation energy for precipitation may be obtained by plotting the logarithm of the time taken to achieve the peak hardness against $1/T$. The results may be fitted by an Arrhenius type equation:

$$\ln(t) = \frac{Q}{RT} + \text{constant} \quad (2)$$

where Q is activation energy for the precipitation (kJ/mol); t the time taken to attain peak hardness at each temperature (h); R the universal gas constant (0.008314 kJ/mol K); T the aging temperature (K).

An activation energy of 132 ± 10 kJ/mol was obtained from the slope of the fitted line (Fig. 5). Using the same method of this work, Viswanathan et al. [14] found an activation energy of 164 kJ/mol in the maraging 350 and Zhang et al. [15] found 85.8 kJ/mol in the maraging 250. Table 3 shows the comparison of these data and presents the base

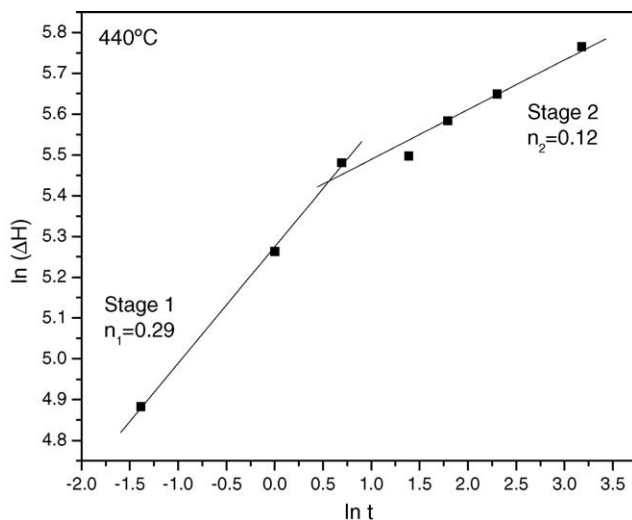


Fig. 4. $\ln(\Delta H)$ vs. $\ln(t)$ for the samples aged at 440 °C using two fitting lines.

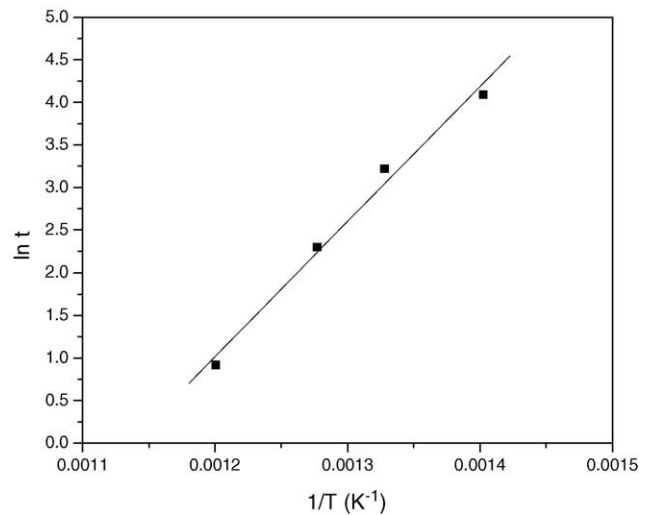


Fig. 5. Plot of $\ln(t)$ vs. $1/T$ for determination of the activation energy for precipitation.

chemical composition of each grade. The three steels present about the same nickel (18% Ni), but differ in the Co, Mo and Ti contents. The Ti/Mo ratio and the cobalt content increase from the 250 to the 350 grades. These modifications enhance the precipitation hardenability providing higher strength levels and aging peaks, but increase the activation energy for precipitation.

Wilson [10] and Sha [11] have also proposed a model for overaging based on the Orowan mechanism:

$$\left(\frac{1}{\Delta H}\right)^3 = M(t - t_0) + \left(\frac{1}{\Delta H_0}\right)^3 \quad (3)$$

where ΔH is difference between the hardening of the overaged sample and the solution treated one (HV); M the temperature dependent rate constant; t_0 the time at peak hardness (h); ΔH_0 the increase in hardness at commencement of coarsening time t_0 (HV).

Fig. 6 shows the precipitation hardening curves of the maraging 300 steel aged at 560, 600 and 650 °C and Fig. 7 shows the fittings $(\Delta H)^{-3}$ versus aging time at 560 and 600 °C used to obtain the parameters M and ΔH_0 presented in Table 4. The correlation coefficients (R) of the linear fittings are also shown. The comparison between the ΔH_0 obtained by the fitting with the experimental value also gives an idea of the validity of the model. The work of Sha [11] shows the application of Eq. (3) on the over-

Table 3
Activation energies for precipitation

Maraging steels	Base chemical composition (wt.%)	Q (kJ/mol)	Reference
250	18Ni–8.5Co–5.0Mo–0.4Ti–0.1Al	85.8	Zhang et al. [15]
300	17.9Ni–9.4Co–4.8Mo–0.8Ti–0.1Al	132	This work
350	18.4Ni–12.3Co–4.0Mo–1.6Ti–0.1Al	164.0	Viswanathan et al. [14]

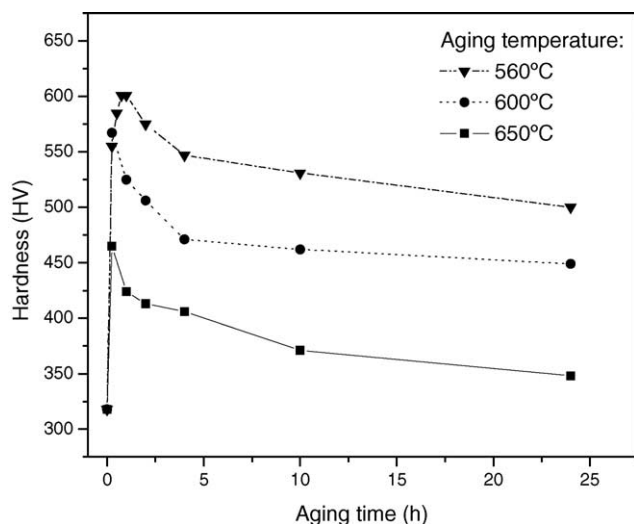
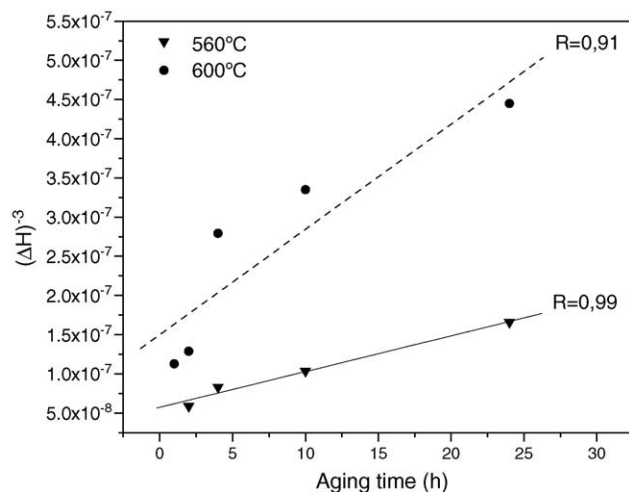


Fig. 6. Hardness vs. aging time at 560, 600 and 650 °C.

Fig. 7. Plots of $(\Delta H)^{-3}$ vs. aging time at 560 and 600 °C for modeling of the overaging kinetics.

aging of a Cr-containing maraging steel at 520 °C. At this temperature, he obtained a good correlation coefficient ($R^2 = 0.98$) with the modeled ΔH_0 (140HV) very close to the experimental value (141HV). In the present work, the model applied at 560 °C presents satisfactory results, with $R = 0.99$ ($R^2 = 0.98$) and ΔH_0 (253HV) close to the experimental value (257HV). The modeling at 600 and 650 °C does not give good results. At 600 °C, the correlation coefficient is very low ($R = 0.91$) and the determined ΔH_0 value (184 HV) is very different from the experimental value (207HV). The fitting at 650 °C (not shown) gives an

Table 4
 M and ΔH_0 values at 560 and 600 °C

Temperature (°C)	M	ΔH_0 (model)	ΔH_0 (experimental)	R
560	4.57E-09	253	257	0.99
600	1.30E-08	184	207	0.91

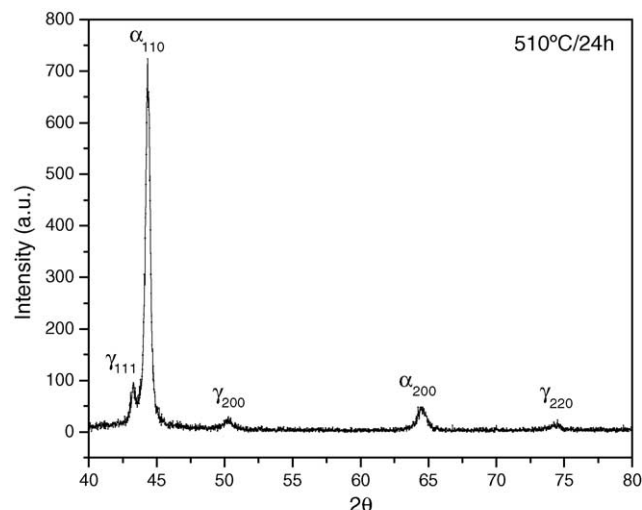


Fig. 8. X-ray diffractogram of the sample aged at 510 °C for 24 h.

$R = 0.97$ and a negative ΔH_0 , which has no physical meaning.

The overaging of the maraging steel aged in temperatures higher than 500 °C is enhanced by the austenite formation. Fig. 8 shows the X-ray diffractogram of the sample aged at 510 °C for 24 h with austenite peaks. However, despite this, satisfactory fittings are obtained at 520 [11] and 560 °C (this work). Fig. 9 shows that the amounts of austenite formed at 600 and 650 °C are much higher than at 560 and 510 °C, which seems to be the reason, why Eq. (3) does not describe well the overaging at 600 and 650 °C.

On the other hand, the overaging of the maraging 300 steel at 560, 600 and 650 °C can be mathematically described by an equation of the type (Fig. 10):

$$H(t) = H_1 t^p$$

where H is hardness value (HV); t the aging time (h); H_1 the hardness value at $t = 1$ h; p the negative exponent of time.

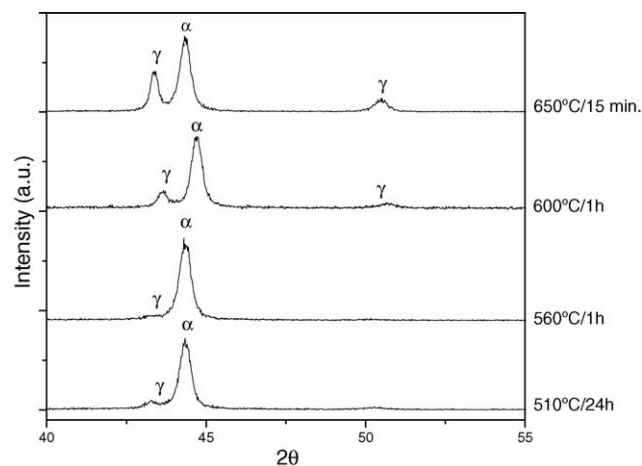


Fig. 9. Comparison between the X-ray diffractograms of the samples aged at 510 °C for 24 h, 560 °C for 1 h, 600 °C for 1 h and 650 °C for 15 min.

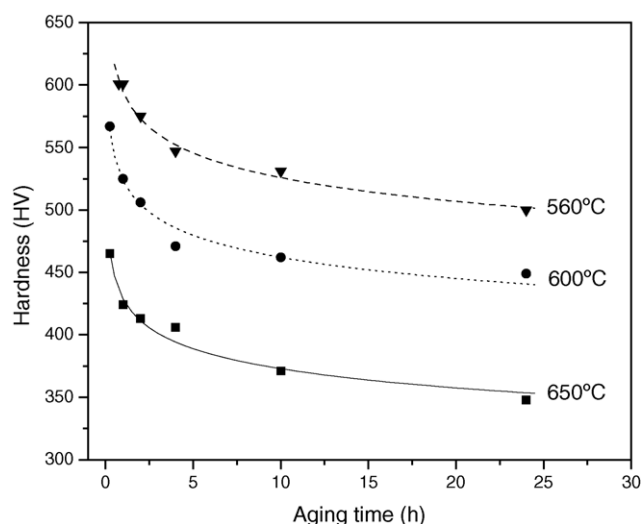


Fig. 10. Fittings of the hardness vs. aging time curves at 560, 600 and 650 °C in the overaged portion.

Table 5
 H_1 and p values at 560, 600, 650 °C

Temperature (°C)	p	H_1 (model)	H_1 (experimental)	R
560	−0.053	594.3	601	0.99
600	−0.054	523.3	525	0.98
650	−0.061	428.8	424	0.99

Fig. 10 shows the curves fitted for 560, 600 and 650 °C. Table 5 shows the parameters of the equations (H_1 and p) and the correlation coefficients obtained. The H_1 obtained by the model and the experimental values (also shown in Table 5) are very close, which validates the model.

4. Conclusions

The age hardening of the maraging 300 steel between 440 and 560 °C can be modeled by the equation $\Delta H = (kt)^n$ with good correlation coefficients (R). The n values found at 440, 480 and 510 °C were 0.20, 0.20 and 0.21, respectively. The n value falls to 0.16 at the aging temperature of 560 °C, which is attributed to the influence of austenite formation during the aging at this temperature. The aging at 440 °C could also be modeled by two curves, representing different stages of precipitation.

An activation energy for precipitation hardening of 132 ± 10 kJ/mol was calculated by the Arrhenius equation. This value is between the ones obtained in maraging 250 [15] and maraging 350 [14] steels.

The model proposed by Wilson [10] for overaging was tested for the samples aged at 560, 600 and 650 °C, but was only valid at 560 °C. The overaging at 650, 600 °C is rather due to the softening effect of the intense austenite formation, and for this reason, is not described by the equation proposed by Wilson [10]. On the other hand, the overaging at 560, 600 and 650 °C was well described by an equation $H = H_1 \cdot t^p$, where H_1 is the hardness value for $t = 1$ h and p is a negative exponent.

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